S ShelfReady

The trust layer for product data

Gemma 4 31B on AMD checks supplier data against its own photos, labels and drawings, asks a person only what the evidence can't settle, and publishes EU-ready listings in 8 languages from verified facts only.

Reinvent Commerce | Gemma 4 31B on AMD | Team Zenin

THE PROBLEM

Supplier data contradicts itself, and the shop pays for it

60.4%

of German online shoppers have returned items because the description or photos did not match (KPMG/EHI 2025)

GPSR Art. 19

Since 13 Dec 2024 every EU online offer must show the manufacturer, an EU responsible person and safety warnings in the shopper's language

76%

of shoppers prefer their own language (CSA 2020); small shops copy supplier PDFs by hand into 5–8 languages

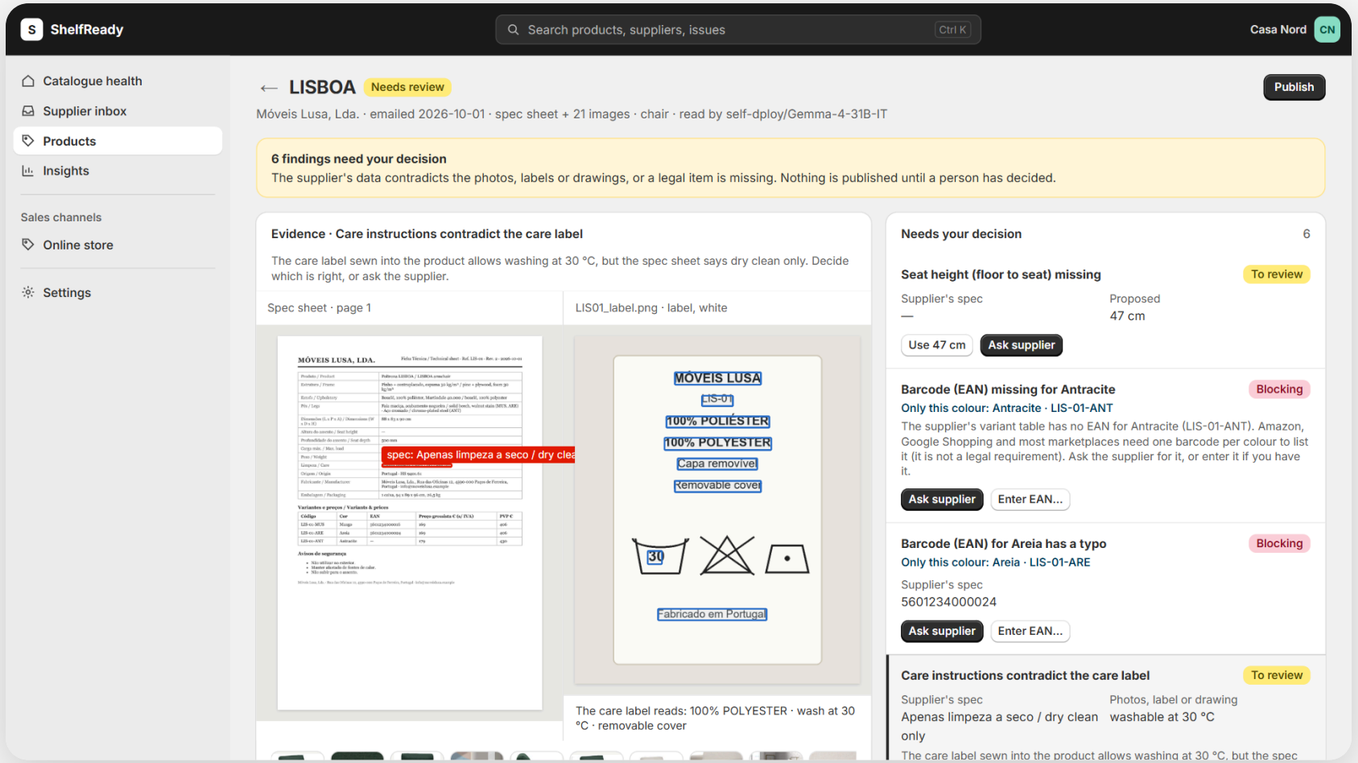
A spec sheet says "dry clean only", the sewn-in label says "wash at 30 °C". The spec says chrome legs, the photos show wood. Nobody compares them before the product goes live.

WHAT IT DOES

From a supplier email to a verified, multilingual listing

1. **Reads the email:** spec sheet PDF, photos, care label, dimension drawing
2. **Gemma 4 31B on AMD** extracts the spec and reads every photo, label and drawing, with boxes
3. **17 checks in Python** compare the supplier's claims with its own evidence and EU rules; units in code
4. **Settles what evidence settles**, asks a person the rest, with both sources side by side
5. **Writes the listing from verified facts only**: copy, safety warnings, alt text for every photo
6. **Translates into 7 languages** and checks numbers, units, names and glossary in code
7. **Publish gate:** nothing goes live with an open legal item or supplier question
8. **Catalogue health:** the same checks audit the shop's existing listings

EVIDENCE, NOT GUESSES



Both sources side by side, the contradiction highlighted

The spec sheet says **dry clean only**; the care label sewn into the chair allows **washing at 30 °C**. A person decides in seconds, and the decision is logged.

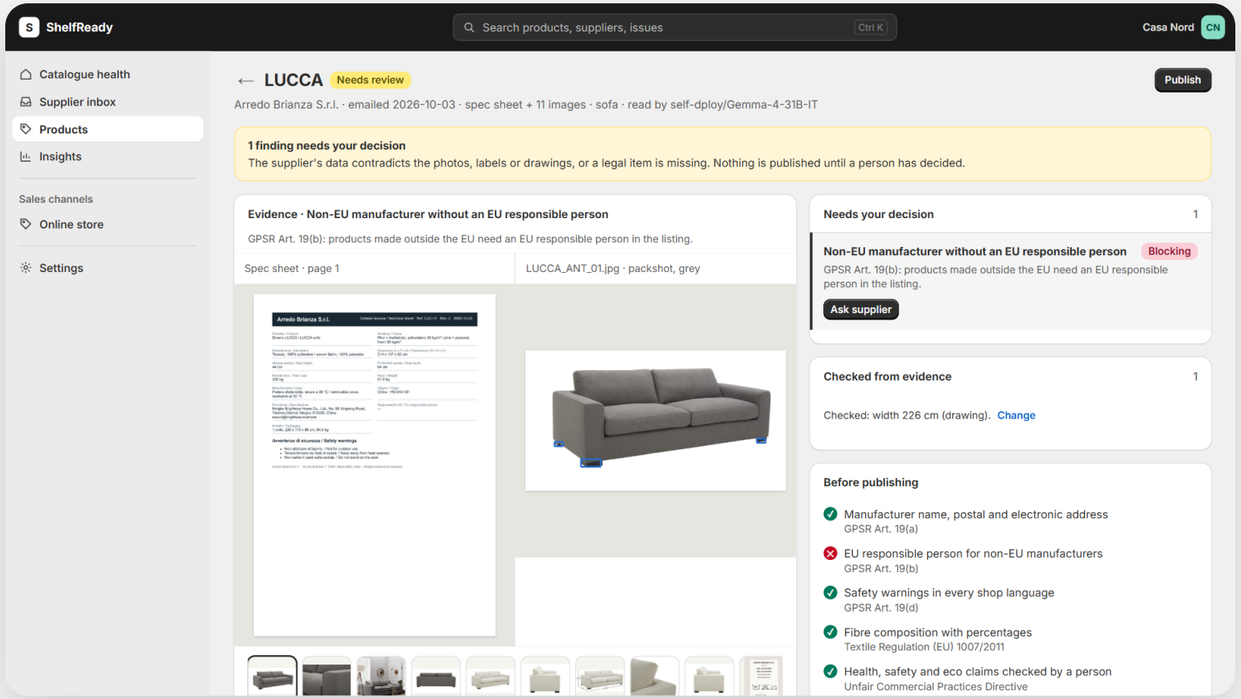
Open facts are held back from the copy instead of guessed.

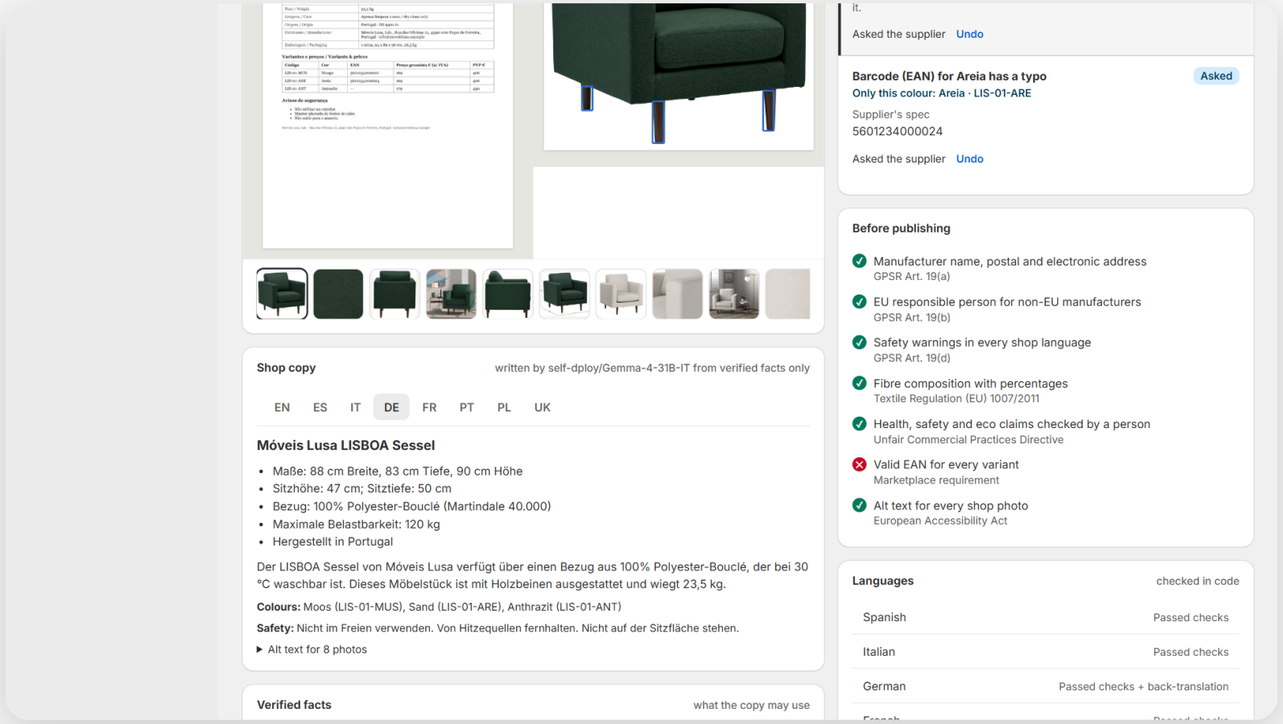
ASK ONLY WHAT THE EVIDENCE CAN'T SETTLE

Settled from evidence: one line. The rest: a clear question.

The supplier's own drawing shows 226 cm, so the width is settled ("Checked: width 226 cm (drawing)") and can still be changed. Photo readings such as colour, and legal gaps, always go to a person.

The **publish gate** stays closed while a legal item or a supplier question is open: GPSR Art. 19, textile fibres, claims, EAN, alt text.





8 languages, every translation checked in code

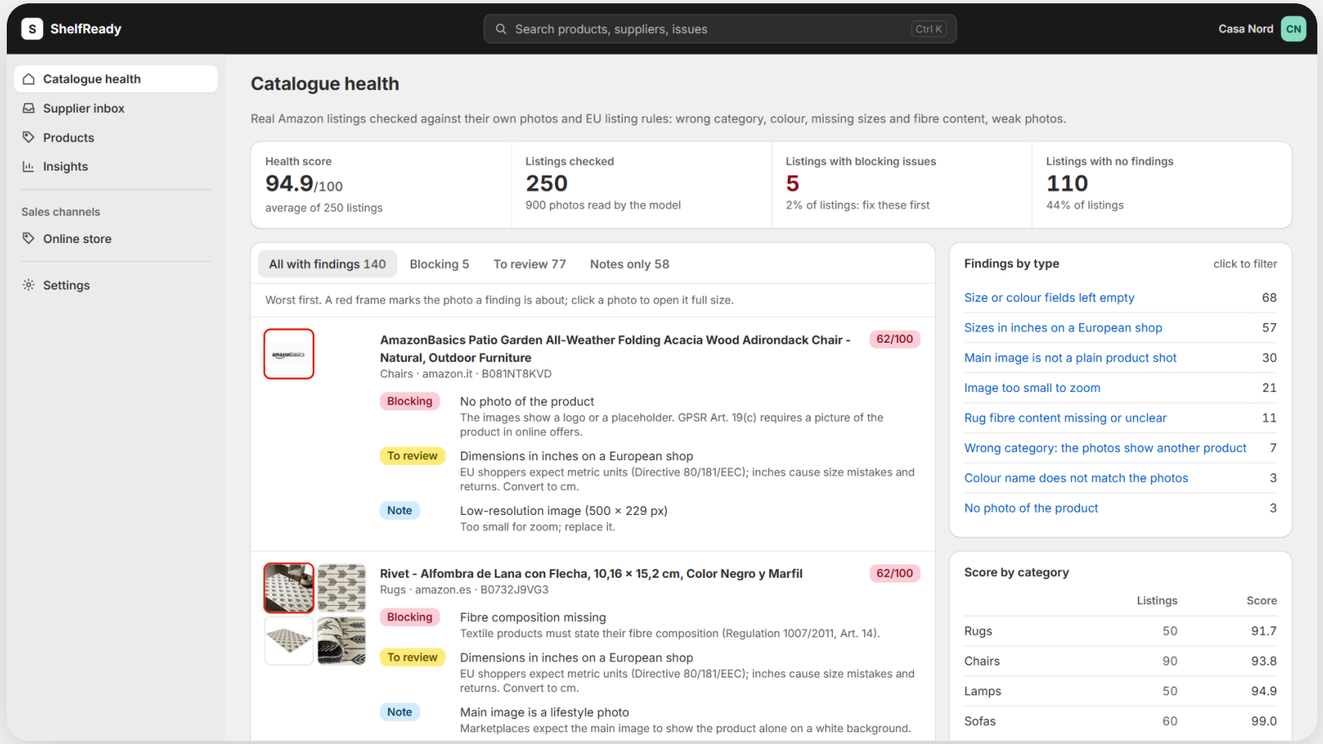
- Title, bullets, description, safety warnings, alt text for every photo
- EN → ES, IT, DE, FR, PT, PL, UK
- Numbers, units, protected names and glossary checked in code, not by the model

CATALOGUE HEALTH ON REAL LISTINGS

250 real listings, 900 photos audited

Amazon Berkeley Objects (CC BY 4.0), checked against their own photos.

- a wristwatch listed under chairs
- listings whose only photo is a logo or "Image coming soon"
- a "Marrone" (brown) sofa whose photos are grey



MEASURED, NOT CLAIMED

Results on real Gemma 4 31B output

57 / 59
planted defects found in 25 supplier submissions (recall 0.966)

0.983
precision: 57 of 58 flags were real problems

0 / 6
clean products falsely flagged

94.9
health score of 250 real listings; 5 with blocking issues

29 / 30
audit photo findings judged right by a person (97%)

8
shop languages, each translation checked in code

Ground truth: data/defects.yaml · method in the repo README

BUILT ON AMD

Gemma 4 31B, the largest Gemma 4, on AMD

- **Every model call runs on AMD:** Gemma-4-31B-IT on **AMD Token Factory** extracts specs, reads photos, labels and drawings, writes and translates the copy
- **Served ourselves on an AMD Instinct MI300X** (AMD Developer Cloud, vLLM on ROCm): 10/10 spec fields, accurate boxes, tool calls
- 1,480+ real answers recorded, so the demo replays real model output

144

photos per minute on one MI300X at 64 concurrent requests

€0.21

per 1,000 photos at $1.99/h

Supplier email → Gemma 4 31B on AMD → 17 checks in Python → Person decides → Gemma 4: copy + 7 languages →

Checks in code → Publish gate

S ShelfReady

Fewer returns. Compliant listings in every language. A person in control of every fact.

Next: supplier questions sent by email, export to Shopify and PIMs, more categories.

shelfready-liard.vercel.app

github.com/KostiantynBk/shelfready